Residual certificates for the Erdős–Straus conjecture: exact solvability criteria, a sieve bound, and computations to 10^{10}

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Abstract

The Erdős–Straus conjecture asserts that $4/n = 1/a + 1/b + 1/c$ is solvable in positive integers for every $n \geq 2$. The open cases concentrate on primes in the six residue classes $\{1, 121, 169, 289, 361, 529\}$ mod 840 (*hard primes*). We study the *residual method*: for $R \equiv 3 \pmod 4$ set $a = (p + R)/4$, $m = pa$; a solution with first denominator a exists if and only if some divisor $k \mid m^2$ satisfies $k \equiv -m \pmod R$. Our main theoretical results are: (i) *exact* solvability criteria for the residuals $R = 3$, 7, and 11 at hard primes, the latter two established by machine-verified finite case analyses; (ii) a meta-theorem showing such a finite exact criterion exists and is computable for every fixed R; (iii) an unconditional chain of sieve bounds: the number of hard primes $p \leq x$ solved by no residual in $\{3, 7\}$, $\{3, 7, 11\}$, $\{3, 7, 11, 19\}$, $\{3, 7, 11, 19, 23\}$ is respectively $O\big(x/(\log x)^2\big)$, $O\big(x/(\log x)^{5/2}\big)$, $O\big(x/(\log x)^3\big)$, $O\big(x/(\log x)^{7/2}\big)$, the last two via a machine-verified *support-bound lemma* that makes every failure at $R = 19, 23$ a sifting condition of dimension $\geq \frac{1}{2}$; (iv) a general density reduction: for every A there is a finite residual list whose joint exceptional set has relative density $O((\log x)^{-A})$ among hard primes. Computationally, we exhibit verified residual certificates for all 14,215,707 hard primes below 10^{10}: the maximal minimal residual is 107, attained at the single prime 8,803,369 and unchanged from 5×10^7 to 10^{10}; a fixed list of 18 residuals covers every hard prime below 10^9. The measured density of the joint $\{3, 7\}$-exceptional set is $\approx 5.17/\log x$, matching the sieve exponent exactly.

1 Introduction

Erdős and Straus conjectured that for every integer $n \geq 2$ the equation

$$\frac{4}{n} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \tag{1}$$

has a solution in positive integers a, b, c. By multiplicativity it suffices to treat prime n, and classical polynomial identities settle every prime outside the six residue classes

$$\mathcal{H} \bmod 840 = \{1,\ 121,\ 169,\ 289,\ 361,\ 529\},$$

the squares of units modulo 840 (Mordell [4]). We call primes in these classes *hard primes*. Direct verification has confirmed the conjecture far beyond any range considered here (to 10^{14}

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by Swett, to 10^{17} by Salez [5], and further in recent work[1]); the structure of the solution count was analyzed by Elsholtz and Tao [1]. Verification, however, exhibits no mechanism; the aim here is a quantitative structural reduction.

1.1 The residual method

Fix a hard prime p. All six classes satisfy

$$\text{(H1) } p \equiv 1 \pmod 8, \qquad \text{(H2) } p \equiv 1 \pmod 3, \qquad \text{(H3) } p \bmod 7 \in \{1, 2, 4\}. \tag{2}$$

For an *admissible residual* $R \equiv 3 \pmod 4$ put

$$a = \frac{p+R}{4} \in \mathbb{Z}, \qquad m = p\,a.$$

Substituting a into (1) and clearing denominators shows that a solution with first denominator a exists if and only if

$$\exists\, k \mid m^2 \quad \text{with} \quad k \equiv -m \pmod R, \tag{3}$$

in which case $b = (k+m)/R$ and $c = (m^2/k + m)/R$ are positive integers solving (1). We write $R_{\min}(p)$ for the least admissible R for which (3) holds. Two elementary congruences drive everything below. From $4a = p + R$,

$$a \equiv 4^{-1}p, \qquad m \equiv 4^{-1}p^2 \pmod R, \tag{4}$$

and since $\gcd(p, R) = 1$ (as $p > R$ is prime), also $\gcd(a, R) = \gcd(m, R) = 1$: the setting is always a unit computation modulo R. By (4), m is a *square* modulo every prime $r \mid R$; hence for every prime $r \mid R$ with $r \equiv 3 \pmod 4$ (at least one exists, since $R \equiv 3 \pmod 4$),

$$\left(\tfrac{-m}{r}\right) = \left(\tfrac{-1}{r}\right) = -1 : \tag{5}$$

the target class $-m$ is a quadratic non-residue mod r.

1.2 Main results

Theorem 1.1 ($R = 3$; exact)**.** *Let $p \equiv 1 \pmod 4$ with $3 \nmid p$, and $a = (p+3)/4$. Then* (3) *holds for $R = 3$ if and only if a has a prime factor $q \equiv 2 \pmod 3$. When it does, $k = q$ is a certificate.*

Theorem 1.2 ($R = 7$, hard primes; exact)**.** *Let p be a hard prime and $a = (p+7)/4$. Then* (3) *holds for $R = 7$ if and only if a has a prime factor that is a quadratic non-residue modulo 7.*

Theorem 1.3 ($R = 11$, hard primes; exact)**.** *Let p be a hard prime and $a = (p+11)/4$; note $3 \mid a$, $2 \nmid a$, $7 \nmid a$ automatically. Then* (3) fails *for $R = 11$ if and only if either*

(a) every prime factor of a is a quadratic residue mod 11; or

(b) $v_3(a) = 1$, every other prime factor of a is $\equiv 1 \pmod{11}$ except a non-residue part of exactly one of the shapes

$$\{q \equiv 2\}, \qquad \{q \equiv 6\}, \qquad \{q \equiv 2\} \cup \{q' \equiv 6\} \pmod{11},$$

each with multiplicity one, paired respectively with $p \equiv 2,\ 6,\ 1 \pmod{11}$.

[1] Verification to 10^{18} is reported in recent literature; the present paper is logically independent of that bound.

Theorems 1.2 and 1.3 are established by finite, machine-verified case analyses (Section 3); the reduction to a finite check is itself a theorem:

Remark 1.4 (computer-assisted components). The exhaustive verifications behind Theorems 1.2, 1.3 and Lemma 1.7 below are *parts of the respective proofs*, on the same footing as a hand case analysis. Each is a deterministic enumeration of a finite configuration space whose finiteness and sufficiency are proved in advance (Theorem 1.5; for Lemma 1.7, monotonicity of target-reachability in support and multiplicities, checked at minimal exponents). The checks are short (seconds on commodity hardware), involve only exact integer arithmetic, and the code is available in the accompanying repository; Appendix A records the exact configuration counts and entry points.

Theorem 1.5 (meta-theorem). *For every fixed prime $R \equiv 3 \pmod 4$, the validity of* (3) *at a hard prime p depends only on* (i) *the multiset of classes mod R of the prime factors of $a = (p+R)/4$, with multiplicities capped at $\lceil (R-1)/2 \rceil$, and* (ii) $p \bmod R$. *Consequently an exact solvability criterion exists for every R and is computable by finite enumeration.*

The anchor result converts the exact criteria into an unconditional bound.

Theorem 1.6 (sieve bound). *The number of hard primes $p \le x$ with $R_{\min}(p) \ge 11$ (that is, solved by neither $R = 3$ nor $R = 7$) is $O\big(x/(\log x)^2\big)$.*

The exponent grows by $\frac{1}{2}$ for each further residual with a sieve-grade handle. For $R = 11$ this handle is Theorem 1.3; for $R = 19, 23$ it is the following machine-verified lemma (Section 4).

Lemma 1.7 (support bound; verified for $R = 19, 23$). *Let $R \in \{19, 23\}$. Every configuration for which residual R fails at a hard prime has at most $(R-3)/2$ nonzero factor-class support; equivalently, the prime factors of $(p+R)/4$ lie in at most $(R-1)/2$ of the $R-1$ unit classes mod R. Consequently failure at R forbids at least half the classes, through finitely many maximal-support branches.*

Theorem 1.8 (the chain). *The number of hard primes $p \le x$ failing every residual in $\{3, 7, 11\}$, in $\{3, 7, 11, 19\}$, and in $\{3, 7, 11, 19, 23\}$ is, respectively,*

$$O\left(\frac{x}{(\log x)^{5/2}}\right), \qquad O\left(\frac{x}{(\log x)^{3}}\right), \qquad O\left(\frac{x}{(\log x)^{7/2}}\right).$$

In particular all but a proportion $O((\log x)^{-2})$ of hard primes are solved within $\{3, 7, 11, 19\}$.

Theorem 1.9 (density reduction). *For every $A > 0$ there is a finite admissible set $S(A)$ such that the number of hard primes $p \le x$ for which every residual in $S(A)$ fails is $O_A\big(x/(\log x)^{1+A}\big)$.*

Computations (Section 5) provide verified certificates for all hard primes below 10^{10} and calibrate every constant in sight.

2 The obstruction framework

Proposition 2.1 (character obstruction). *Let $r \mid R$ be prime with $r \equiv 3 \pmod 4$. If every prime factor q of $m = pa$ satisfies $\left(\frac{q}{r}\right) = +1$, then* (3) *fails for R.*

Proof. Every divisor k of m^2 is a product of prime factors of m, so $\left(\frac{k}{r}\right) = +1$. A certificate requires $k \equiv -m \pmod r$, whence $\left(\frac{k}{r}\right) = \left(\frac{-m}{r}\right) = -1$ by (5) — impossible. □

We call failures produced by Proposition 2.1 *Type I*; failures where m possesses a non-residue prime factor modulo every relevant r, yet no divisor of m^2 attains the exact class $-m \bmod R$, are *Type II*.

Proposition 2.2 (guaranteed-success classes)**.** *Let* $t \equiv -4^{-1}p^2 \pmod{R}$*. If* a *has a prime factor* q *with* $q \equiv t\,p^{-i} \pmod{R}$ *for some* $i \in \{0,1,2\}$*, then* $k = q\,p^i$ *is a certificate. In particular, since* $t\,p^{-2} \equiv -4^{-1} \pmod{R}$*, any prime factor of* a *in the fixed class* $-4^{-1} \bmod R$ *certifies success regardless of* p*.*

Proof. $k = qp^i$ divides m^2 (as $q \mid a$ and $i \leq 2$) and lies in class $t \equiv -m$. Integrality of b, c is automatic from $k \equiv -m$ and $m^2/k \equiv m^2(-m)^{-1} \equiv -m \pmod{R}$. □

The contrapositive of Proposition 2.2 is the sieve input for Theorem 1.9: failure at R forces $(p+R)/4$ to avoid prime factors in up to three explicit classes modulo R.

3 Exact criteria

3.1 Proof of Theorem 1.1

Modulo 3 the unit group has order 2, so quadratic character equals class. Since $p^2 \equiv 1 \pmod{3}$, (4) gives $m \equiv 1$ and the target is $-m \equiv 2 \pmod{3}$; also $3 \nmid a$. *Necessity:* if every prime factor of a is $\equiv 1 \pmod{3}$, then $a \equiv 1 \pmod{3}$, hence $p = 4a - 3 \equiv 1 \pmod{3}$ as well, so every prime factor of $m = pa$ is $\equiv 1 \pmod{3}$ and every divisor of m^2 is $\equiv 1 \pmod{3}$: class 2 is never attained. *Sufficiency:* $k = q \equiv 2 \pmod{3}$ divides m^2, and the recovery formulae give integers as in Proposition 2.2. □

3.2 Proof of Theorem 1.5

A certificate is a divisor $k \mid m^2$ in class $-m \bmod R$. The set of classes of divisors of m^2 is determined by the classes of the prime factors of $m = pa$ together with their multiplicities, each factor being usable with exponent at most twice its multiplicity; exponents are only meaningful modulo $\mathrm{ord}_R(\text{class}) \leq R-1$, so multiplicities beyond $\lceil (R-1)/2 \rceil$ are equivalent to that cap. The target class and the consistency relation $\prod(\text{classes}) \equiv a \equiv 4^{-1}p \pmod{R}$ are functions of $p \bmod R$ by (4). Hence solvability is a function of the capped class multiset and $p \bmod R$, a finite amount of data. □

3.3 Proof of Theorem 1.2

($\Rightarrow$) is Proposition 2.1 with $r = R = 7$. ($\Leftarrow$) reduces to a finite verification via Theorem 1.5. Write $(\mathbb{Z}/7)^* \cong \mathbb{Z}/6$ in discrete-log coordinates. Three structural constraints hold at hard primes:

(i) $p \bmod 7 \in \{1, 2, 4\}$, by (H3);

(ii) $2 \mid a$: by (H1), $p + 7 \equiv 0 \pmod{8}$, so $a = (p+7)/4$ is even — the class of 2 (a residue mod 7) is always present;

(iii) the factor classes multiply to $a \equiv 4^{-1}p \pmod{7}$.

Enumerating all class multisets with multiplicities capped at 3 subject to (i)–(iii) yields 1536 consistent configurations; for each, one checks whether the target class is attained by a bounded-exponent subproduct. The verification (code: `theory.verify_R7_finite`) confirms: the target is attainable precisely when some factor class is a non-residue. □

Remark 3.1. The hard-class hypotheses are essential: for $p = 701 \equiv 1 \pmod 4$ (not hard), $a = 177 = 3 \cdot 59$ has the non-residue factor 3, yet $R = 7$ fails. Constraint (ii) — the forced even factor — is exactly what rescues every would-be Type II configuration at hard primes. Empirically (Section 5), among 41,421 sampled failures of $R = 7$ at hard primes, *all* are Type I.

3.4 Proof of Theorem 1.3

By Theorem 1.5 with $R = 11$, $d = 10$: multiplicities cap at 5, and at hard primes $3 \mid a$ (from (H2), $11 \equiv 2 \bmod 3$), while $2 \nmid a$ (from (H1), $p + 11 \equiv 4 \bmod 8$) and $7 \nmid a$ (from (H3), $-11 \equiv 3 \bmod 7 \notin \{1, 2, 4\}$). A dynamic-programming enumeration over the finite configuration space (code: `theory.finite_criterion_dp`) compresses it to 25 equivalence states: 16 succeed, 6 fail with all factor classes quadratic residues (case (a)), and exactly 3 fail with a non-residue present (case (b)); the three failing states decode to the three listed patterns. Two structural facts emerge from the enumeration. First, no state fails by a *proper-subgroup* obstruction: the only subgroup of $\mathbb{Z}/10$ containing an odd element properly is $\{0, 5\}$, and the consistency relation forces p's class to extend it to all of $\mathbb{Z}/10$. Second, the case-(b) failures are pure exponent-budget edge cases: any of $v_3(a) \geq 2$, a factor in class $\{4, 5, 9\}$, or a second non-residue factor restores solvability. The criterion agrees with ground-truth divisor search for 158,759 of 158,759 hard primes tested below 10^9. □

4 The sieve bound

Proof of Theorem 1.6. Step 1 (exact criteria). Let p be a hard prime and set $n = (p+3)/4$, so that $(p+7)/4 = n+1$: *the two shifted forms are consecutive integers.* By Theorems 1.1 and 1.2, p is exceptional (both residuals fail) precisely when

$$n \text{ has no prime factor} \equiv 2 \pmod 3 \quad \text{and} \quad n+1 \text{ has no prime factor} \equiv 3, 5, 6 \pmod 7,$$

with $p = 4n - 3$ prime.

Step 2 (congruence bookkeeping). Fix one of the six hard classes $h \bmod 840$; then $n \equiv (h+3)/4 \pmod{210}$ is fixed, and the conditions at the primes $2, 3, 5, 7$ are determined by the class (in particular $3 \nmid n$ and $7 \nmid n+1$ automatically). It suffices to bound each class and sum.

Step 3 (containment in a sifted set). Let $z = x^{1/10}$. After discarding the $O(z)$ primes $p \leq z$, every exceptional p gives $n \leq (x+3)/4$, $n \equiv n_0 \pmod{210}$, such that for every prime q with $7 < q \leq z$:

(a) $q \equiv 2 \pmod 3 \implies n \not\equiv 0 \pmod q$;

(b) $q \equiv 3, 5, 6 \pmod 7 \implies n \not\equiv -1 \pmod q$;

(c) $q \nmid 4n - 3$ (since $4n - 3 = p$ is prime, $p > z$).

Conditions (a), (b) only use "no forbidden factor up to z", which is implied by the full criteria — an upper bound needs no more.

Step 4 (sieve of dimension 2). This is a standard upper-bound sieve for the three linear forms n, $n+1$, $4n-3$, with sifting function

$$\omega(q) \;=\; \mathbf{1}_{q\equiv 2(3)} \;+\; \mathbf{1}_{q\,\mathrm{NQR}\,(7)} \;+\; 1 \qquad (q>7),$$

whose forbidden classes $0, -1, 3\cdot 4^{-1} \bmod q$ are pairwise distinct for $q \nmid 21$. By Dirichlet, ω has average $\kappa = \frac{1}{2}+\frac{1}{2}+1 = 2$ in the Halberstam–Richert sense $\sum_{q\le w}\omega(q)\log q/q = \kappa\log w + O(1)$. The Fundamental Lemma of sieve theory ([3, Thm. 2.5]; [2, Lem. 6.3]) with $z = x^{1/10}$ gives

$$\#E(x) \;\ll\; x \prod_{7<q\le z}\Big(1-\frac{\omega(q)}{q}\Big) \;\asymp\; x\,(\log z)^{-1}(\log z)^{-1/2}(\log z)^{-1/2} \;\asymp\; \frac{x}{(\log x)^2},$$

by Mertens' theorem and its arithmetic-progression form. Summing over the six classes preserves the bound.

□

Proof of Lemma 1.7. Reachability of the target class by a bounded-exponent subproduct is monotone in both the support and the multiplicities of the factor-class configuration. It therefore suffices to check that for every support of size $(R-1)/2$ over the nonzero discrete logs and every class of $p \bmod R$, the target is reachable already at minimal multiplicities (exponent budget 2 per class, p-budget 2). The exhaustive check (code: `theory.verify_support_bound`) covers 437,580 pairs at $R = 19$ and 7,759,752 pairs at $R = 23$, with no failures. The bound is tight: the all-residue (Type I) configurations occupy exactly $(R-3)/2$ nonzero classes. □

Proof of Theorem 1.8. Write $n = (p+3)/4$; the relevant forms are n, $n+1$, $n+2 = (p+11)/4$, $n+4 = (p+19)/4$, $n+5 = (p+23)/4$, and $4n-3 = p$. Joint failure of $\{3,7,11\}$ implies, by Theorems 1.1, 1.2, 1.3: n free of primes $\equiv 2 \pmod 3$; $n+1$ free of non-residues mod 7; and, splitting by the two cases of Theorem 1.3: *branch (a)* — $n+2$ free of the five non-residue classes mod 11 (density $\frac{1}{2}$); *branch (b)* — $n+2$ free of the six classes $\{4,5,7,8,9,10\}$ mod 11 (density $\frac{3}{5}$). Each branch is a four-form sieve problem of dimension $1+\frac{1}{2}+\frac{1}{2}+\frac{1}{2} = \frac{5}{2}$, resp. $1+\frac{1}{2}+\frac{1}{2}+\frac{3}{5} = \frac{13}{5}$. Since $\frac{13}{5} > \frac{5}{2}$, the Type-II branch contributes $O\big(x/(\log x)^{13/5}\big) = o\big(x/(\log x)^{5/2}\big)$: the bound is the maximum of the branch exponents, attained by the character branch (a). The Fundamental Lemma applied to each branch gives $O(x/(\log x)^{5/2})$ in total.

For $\{3,7,11,19\}$, Lemma 1.7 splits failure at 19 into finitely many maximal-support branches, each forbidding an explicit set of at least 9 of the 18 unit classes mod 19 on $n+4$ — dimension $\ge \frac{1}{2}$. Crossing with the branches at 11 leaves finitely many combined branches, each a five-form problem of dimension ≥ 3 (those involving the Type-II case at 11 have dimension $\ge 1+\frac{1}{2}+\frac{1}{2}+\frac{3}{5}+\frac{1}{2} = \frac{31}{10} > 3$ and are again lower-order); the forbidden residues of the five forms are pairwise distinct for $q > 19$, smaller primes being absorbed into the congruence class of n. Summing the finitely many branches gives $O(x/(\log x)^3)$. The $\{3,7,11,19,23\}$ case adds the form $n+5$ with Lemma 1.7 at $R = 23$, one more factor of $(\log x)^{1/2}$. □

Remark 4.1. Each further prime residual $R \equiv 3 \pmod 4$ whose support-bound lemma is verified adds $\frac{1}{2}$ to the exponent; the verification costs $\binom{R-2}{(R-1)/2}$ reachability checks, trivial through $R = 23$ and feasible with bit-parallel code to $R \approx 31$–43. The chain is a concrete instantiation of Theorem 1.9 with explicit, verified constants.

Proof of Theorem 1.9. Fix $A > 0$. Choose $B = B(A)$ so large that

$$\sum_{\substack{3\le R\le B\\ R\equiv 3\,(4)}} \frac{1}{\varphi(R)} \;\ge\; \sum_{\substack{3\le R\le B\\ R\equiv 3\,(4)}} \frac{1}{R} \;>\; A; \tag{6}$$

this is possible since the reciprocal sum over the progression $R \equiv 3 \pmod 4$ diverges, and $\varphi(R) \le R$. Let $S(A) = \{R \equiv 3\,(4) : 3 \le R \le B\}$ and $M = 840 \cdot \prod_{R\in S(A)} R$.

Write $n = (p+3)/4$ as before, so $(p+R)/4 = n + (R-3)/4$ for each $R \in S(A)$: a family of $|S(A)|$ integer linear forms in n with pairwise distinct shifts, together with the form $4n-3 = p$. Fix a residue class $n \equiv n_0 \pmod M$ compatible with p hard; there are finitely many, and it suffices to bound each. Within such a class, for every $R \in S(A)$ the quantities $t \equiv -4^{-1}p^2$ and p^{-1} mod R are constant, so the class set $S_R(p) = \{t, tp^{-1}, tp^{-2}\}$ mod R of Proposition 2.2 is a fixed nonempty set of $t_R = t_R(n_0) \in \{1, 2, 3\}$ classes. By the contrapositive of Proposition 2.2, if every residual in $S(A)$ fails at p, then for each $R \in S(A)$ the form $n + (R-3)/4$ has *no* prime factor in $S_R(p)$; and $4n-3$, being prime and $> z$, has no prime factor $\le z$.

Sift the class $\{n \le (x+3)/4 : n \equiv n_0\ (M)\}$ by the primes $q \le z = x^{1/10}$, $q \nmid M$. A prime q forbids: the class $n \equiv 3\cdot 4^{-1} \pmod q$ (from the primality of $4n-3$), and, for each $R \in S(A)$ with q mod $R \in S_R(p)$, the single class $n \equiv -\frac{R-3}{4} \pmod q$ (a prime factor q of the form $n+(R-3)/4$ in a guaranteed-success class cannot occur). Thus

$$\omega(q) \;=\; 1 \;+\; \#\bigl\{R \in S(A) : q \bmod R \in S_R(p)\bigr\},$$

which is bounded by $|S(A)| + 1$; the forbidden residues of distinct forms are pairwise distinct for $q > B$, since the shifts $(R-3)/4$ differ by less than B and $q \nmid M$. For each R, the density of primes q with q mod $R \in S_R(p)$ is $t_R/\varphi(R)$ by Dirichlet's theorem, so ω has average

$$\kappa \;=\; 1 + \sum_{R\in S(A)} \frac{t_R}{\varphi(R)} \;\ge\; 1 + \sum_{R\in S(A)} \frac{1}{\varphi(R)} \;>\; 1 + A$$

in the Halberstam–Richert sense, by (6). The Fundamental Lemma ([3, Thm. 2.5]) in dimension κ then bounds the sifted count by $O_A\Bigl(x\prod_{q\le z}(1-\omega(q)/q)\Bigr) = O_A(x(\log x)^{-\kappa}) = O_A\bigl(x/(\log x)^{1+A}\bigr)$, and summing over the finitely many classes n_0 mod M preserves the bound. □

Remark 4.2. Theorem 1.9 is the strongest statement the pure sieve can give: density $O((\log x)^{-A})$ for every A, but not emptiness. The conjecture for hard primes is equivalent to a *finite covering hypothesis*: some finite S admits, for every hard prime, a residual $R \in S$ with (3). All computational evidence below supports $S = \{3, 7, 11, \ldots, 107\}$.

5 Computations to 10^{10}

All code and data are in the accompanying repository. Certificates are found by direct integer factorization of a (trial division; the factorization of $m^2 = p^2a^2$ follows) and scanning divisors for the class condition (3); every certificate is re-verified in exact integer arithmetic, $4abc = n(bc + ac + ab)$.

5.1 Data

Every one of the 14,215,707 hard primes $p < 10^{10}$ possesses a residual certificate with $R \le 107$. The minimal residuals distribute as follows (10^{10} data):

R	count	R	count	R	count
3	7 386 405	31	30 222	63	57
7	3 472 188	35	4 775	67	18
11	2 325 588	39	4 307	71	46
15	487 889	43	1 131	75	3
19	322 871	47	1 661	79	11
23	150 828	51	240	83	5
27	27 000	55	263	107	1
		59	198		

Salient features:

- $R_{\max} = 107$, attained by the single prime 8,803,369 $< 10^7$, *unchanged from* 5×10^7 *to* 10^{10} — three orders of magnitude.
- No prime has minimal $R \in \{87, 91, 95, 99, 103\}$. This gap requires no modular explanation: under the calibrated independence model the expected counts there are 0.12, 0.03, 0.05, 0.01, 0.01. Conditioned on failing every $R \le 83$, a prime lands at 87 with probability 0.53 and at 107 with probability 0.017; the record prime is the anomaly, not the gap.
- Among all 1,587,581 hard primes below 10^9, exactly *four* admit a unique working residual ≤ 107 (8,803,369 with 107; 142,361,209 with 59; 287,567,281 with 83; 794,037,841 with 63) — the entire obstruction to a shorter covering list. A greedy cover of the full solvability table uses 18 residuals; a packing argument gives a lower bound of 12.

5.2 Calibration of Theorem 1.6

The proportion of hard primes failing both $R = 3$ and $R = 7$, multiplied by $\log x$, is constant to within 2% across three decades of the 10^9 data:

bin ($p \sim$)	10^6	10^7	6×10^7	2.5×10^8	10^9
density $\times \log x$	5.14	5.19	5.18	5.17	5.16

That is, the joint exceptional density is $\approx 5.17/\log x$: the sieve exponent of Theorem 1.6 is exactly right, and 5.17 estimates the (conjectural) Selberg–Delange constant. The exceptional set is density zero but decays slowly: 26.5% of hard primes below 10^9 fail both residuals; all are settled by deeper residuals of the list. The fitted per-residual decay exponents likewise match the sieve prediction $\kappa = \frac{1}{2}$ where exact criteria exist (e.g. $\hat{\kappa} = 0.539$ for $R = 7$).

5.3 Failure taxonomy

Sampling the complete solvability table at 10^9: failures at $R = 7$ are 100% Type I (as Theorem 1.2 requires); at $R = 11$, 89.7% Type I and 10.3% the budget pattern of Theorem 1.3(b); the Type I share decreases with $\varphi(R)$ (e.g. 51% at $R = 103$), as hitting one specific class among $\varphi(R)$ grows harder while the character dichotomy stays binary.

6 Open problems

1. **Finite covering.** Prove that some finite admissible set S covers all hard primes (Theorem 1.9 gives density $(\log x)^{-A}$ for every A; emptiness is beyond the sieve).

2. **Longer chains.** Verify Lemma 1.7 at $R = 31, 43, 47$ (exponents $4, \frac{9}{2}, 5$) with bit-parallel enumeration, and treat the first composite residual $R = 15$ (coupled conditions mod 3 and mod 5), for which the meta-theorem applies componentwise.

3. **The record prime.** 8,803,369 fails every residual below 107 even allowing $R \leq 400$; its shifted values are anomalously smooth (at $R = 107$, $a = 3^2 \cdot 11^2 \cdot 43 \cdot 47$). A quantitative theory of joint failure for smooth-shift primes would explain why the record is static and bound its future growth.

A Verification details

Each computer-assisted component below is a deterministic, exhaustive enumeration of a finite space whose sufficiency is established in the text; all arithmetic is exact integer arithmetic, each run completes in seconds on commodity hardware, and the entry points named are in the module `erdos_straus.theory` of the accompanying repository.

component	configuration space	count	entry point
Thm. 1.2 ($R = 7$)	class multisets in $(\mathbb{Z}/7)^*$, multiplicities ≤ 3, with the forced even factor, $p \bmod 7 \in \{1, 2, 4\}$, and the consistency relation	1,536	`verify_R7_finite`
Thm. 1.3 ($R = 11$)	dynamic program over $(\mathbb{Z}/11)^*$ with the forced factor 3; collapses to equivalence states (16 succeed, 6 fail by character, 3 by budget)	25 states	`finite_criterion_dp`
Lem. 1.7 ($R = 19$)	all supports of size 9 over the nonzero logs $\times$ all classes of p, at minimal multiplicities	437,580	`verify_support _bound`
Lem. 1.7 ($R = 23$)	all supports of size 11, likewise	7,759,752	`verify_support _bound`

In addition (as sanity checks, not proof components), the criteria of Theorems 1.1–1.3 were compared against ground-truth divisor search on the complete solvability data below 10^9: agreement in 1,587,581/1,587,581 cases for $R = 3$ and 158,759/158,759 sampled cases for $R = 11$, with zero discrepancies.

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